

Drugs for Heart Failure

OVERVIEW

- ✓ Heart failure (HF) is a complex, progressive disorder in which the **heart is unable to pump sufficient blood to meet the needs of the body.**
- ✓ Its cardinal symptoms are dyspnea, fatigue, and fluid retention
- ✓ impaired ability of the heart to adequately fill with and/or eject blood
- ✓ abnormal increases in blood volume and interstitial fluid.

- causes of HF:
 - ✓ atherosclerotic heart disease
 - ✓ hypertensive heart disease
 - ✓ valvular heart disease
 - ✓ congenital heart disease

Compensatory Mechanisms

❑ Chronic activation of the SNS and the RAAS

- remodeling of cardiac tissue, loss of myocytes, hypertrophy, and fibrosis

❑ **pharmacologic intervention**

- to alleviate symptoms, slow disease progression, and improve survival

❑ Therapeutics

- ✓ increasing cardiac contractility
- ✓ reducing preload
- ✓ normalizing heart rate and rhythm

- ✓ ACE inhibitors
- ✓ angiotensin-receptor blockers
- ✓ aldosterone antagonists
- ✓ β -blockers
- ✓ diuretics
- ✓ direct vaso- and venodilators
- ✓ HCN channel blockers
- ✓ inotropic agents
- ✓ B type natriuretic peptide

Provides:

- ✓ reduced myocardial work load
- ✓ decreased extracellular fluid volume
- ✓ improved cardiac contractility
- ✓ reduced rate of cardiac remodeling

PHYSIOLOGY OF MUSCLE CONTRACTION

- Cardiac myocytes are **interconnected in groups** that respond to stimuli as a unit, contracting together whenever a single cell is stimulated.

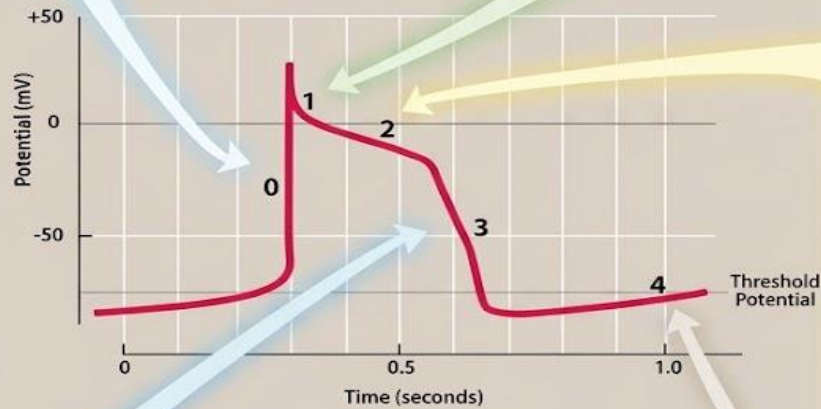
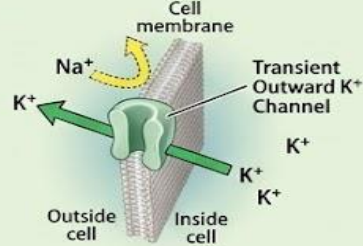
PHASE 0: FAST UPSTROKE

- Na⁺ channels open ("fast channels") resulting in a fast inward current.
- Upstroke ends as Na⁺ channels are rapidly inactivated.
- Sodium current is blocked by anti-arrhythmic agents, such as *quinidine*.



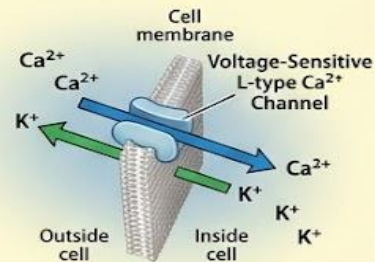
PHASE 1: PARTIAL REPOLARIZATION

- The initial rapid phase of repolarization is due to:
 - 1) Inactivation of Na⁺ channels
 - 2) K⁺ channels that rapidly open and close, causing a transient outward current



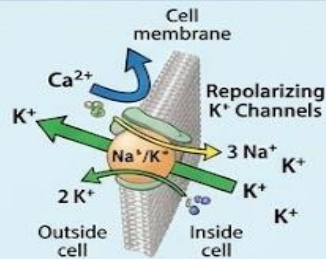
PHASE 2: PLATEAU

- Voltage-sensitive L-type Ca²⁺ Channel open, resulting in a slow inward (depolarizing) current that balances the slow outward (polarizing) leak of K⁺.



PHASE 3: REPOLARIZATION

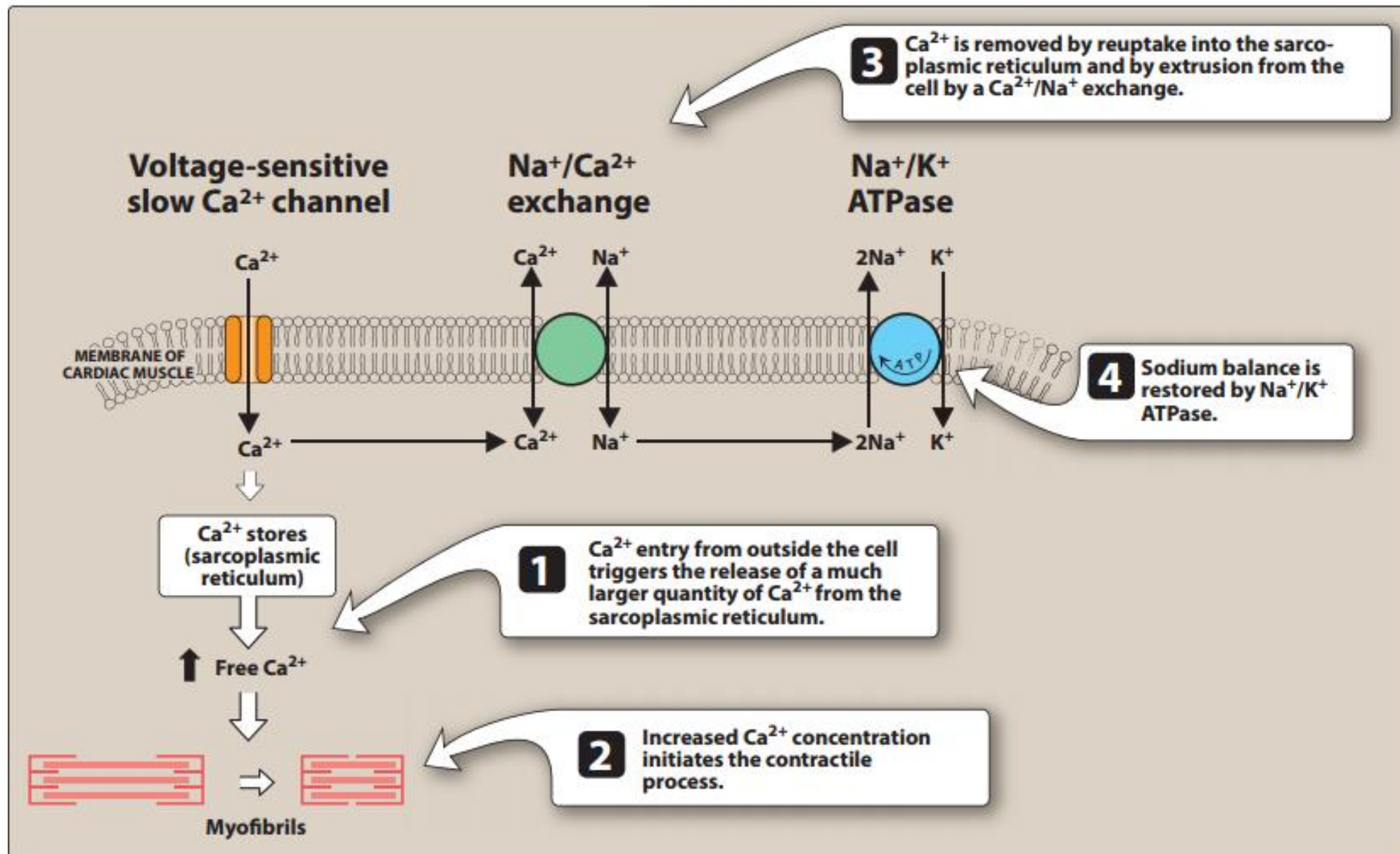
- Ca²⁺ channels close.
- K⁺ channels open, resulting in an outward current that leads to membrane repolarization.
- The net result of the action to this point is a net gain of Na⁺ and loss of K⁺. This imbalance is corrected by Na⁺/K⁺-ATPase.



PHASE 4: FORWARD CURRENT

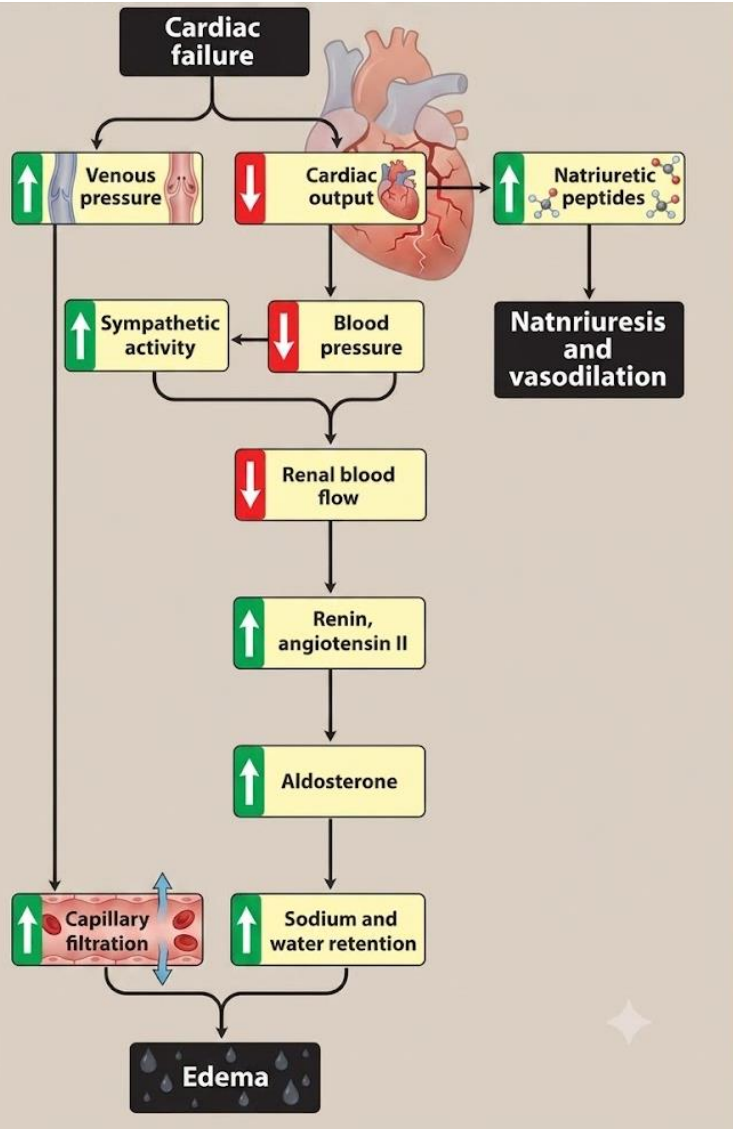
- Increasing depolarization results from gradual increase in sodium permeability.
- The spontaneous depolarization automatically brings the cell to the threshold of the next action potential.

AP of a cardiac myocyte



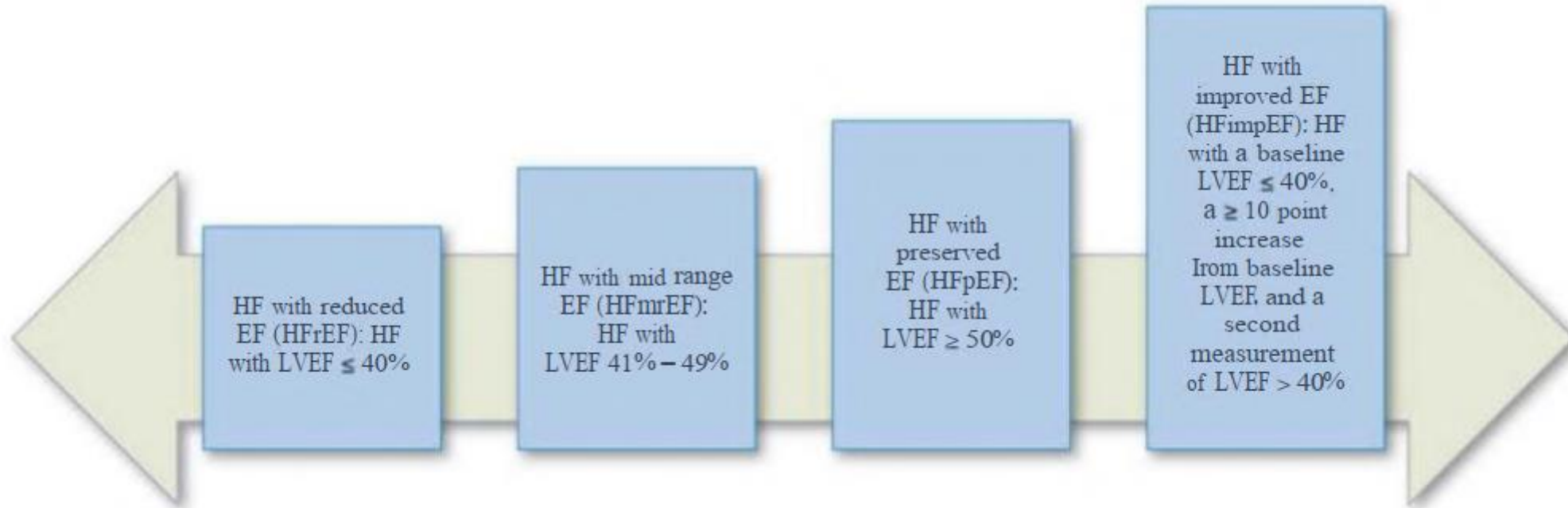
Ion movements during the contraction of cardiac muscle

Compensatory physiological responses in HF

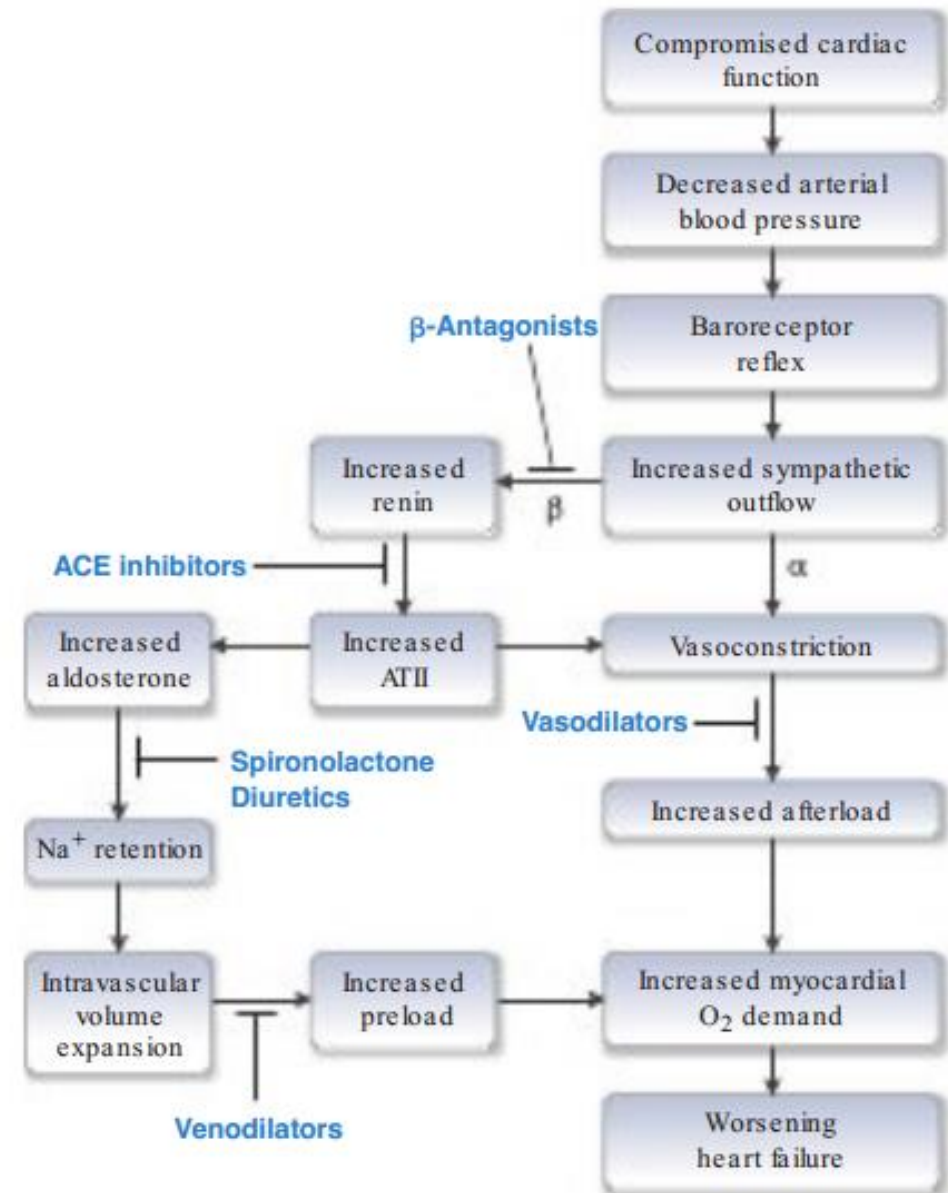
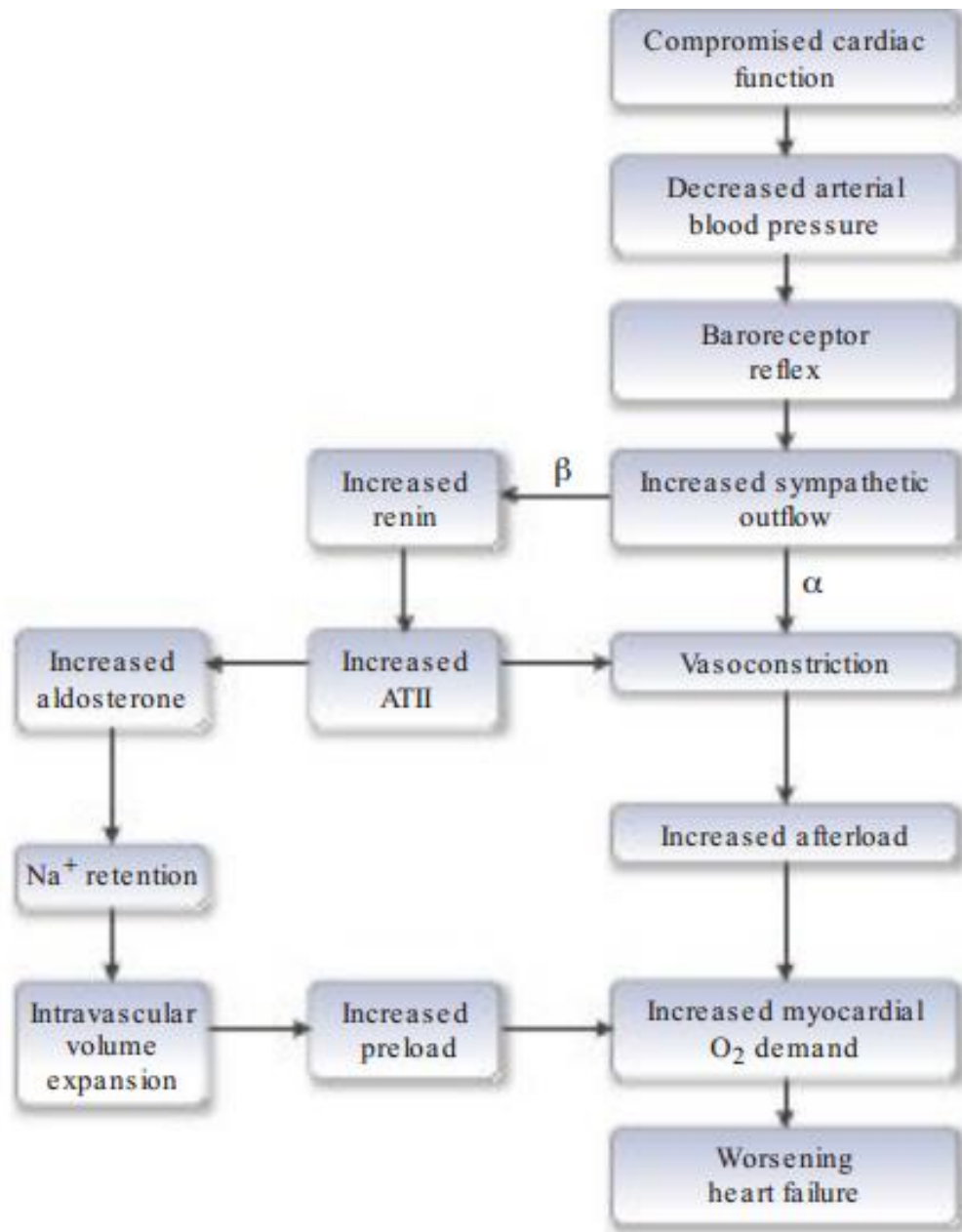


- ✓ Increased sympathetic activity
 - enhances venous return and ↑es cardiac preload (stretch on the heart), ↑es SV, which, in turn, ↑es CO
- ✓ Activation of RAAS
 - ↑preload (PR), BV → ↑VP → edema
- ✓ Activation of natriuretic peptides
- ✓ Myocardial dysfunction
- ✓ favoring remodeling, fbrosis, and inflammatory changes (**ag II and aldosterone**)
- ✓ ↑workload of the heart

Type of Failure	Characteristics	Contributing Factors	Etiology
Low output, systolic dysfunction (dilated cardiomyopathy) ^a	<u>Hypofunctioning left ventricle</u> ; enlarged heart (dilated left ventricle); <u>Tleft ventricular end diastolic volume</u> : EF <40%. <u>Astroke volume</u> : ACO: S, heart sound present	1 <u>AContractility</u> (cardiomyopathy) 2 <u>TAfterload</u> (elevated SVR)	1 <u>Coronary ischemia</u> , ^a MI, mitral valve stenosis or regurgitation, alcoholism, viral syndromes, nutritional deficiency, calcium and potassium depletion, drug-induced, idiopathic 2 <u>Hypertension</u> , aortic stenosis, volume overload
Diastolic dysfunction	<u>Normal left ventricular contractility</u> : normal size heart; stiff left ventricle: <u>impaired left ventricular relaxation</u> ; impaired left ventricular filling: normal EF; S ₄ , heart sound	1 Thickened left ventricle (hypertrophic cardiomyopathy) 2 Stiff left ventricle (restrictive cardiomyopathy) 3 Tpreload	1 Coronary ischemia, ^a MI <u>hypertension</u> , aortic stenosis and <u>regurgitation</u> , pericarditis, enlarged left ventricular septum (hypertrophic cardiomyopathy) 2 Amyloidosis, sarcoidosis 3 <u>Sodium and water retention</u>
High-output failure (uncommon)	Normal or <u>Tcontractility</u> , normal size heart: normal left ventricular end-diastolic volume, normal or <u>TEF</u> , normal or increased stroke volume; TCO	1 <u>tMetabolic and oxygen demands</u>	1 <u>Anemia and hyperthyroidism</u>



Classification of heart failure based on ejection fraction



Cardiac Workload

- increased cardiac workload
- Four major determinants contribute to LV workload

Preload

- ✓ forces acting on the *venous* side of the circulation affecting myocardial wall tension
- ✓ PVD and decreased PVV diminish preload

- ✓ **PV constriction** and **increased PVV**
- ✓ Elevated preload can aggravate HF
- ✓ Rapid administration of blood plasma expanders
- ✓ osmotic diuretics
- ✓ administration of large amounts of sodium or sodium-retaining agents

increase preload

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increase preload

Afterload

- tension developed in the ventricular wall as contraction (systole) occurs
- affected by
 - ✓ intraventricular pressure
 - ✓ ventricular diameter
 - ✓ wall thickness
 - ✓ systemic vascular resistance (SVR)
- the impedance against which the ventricle must pump and is estimated by arterial blood pressure (BP)

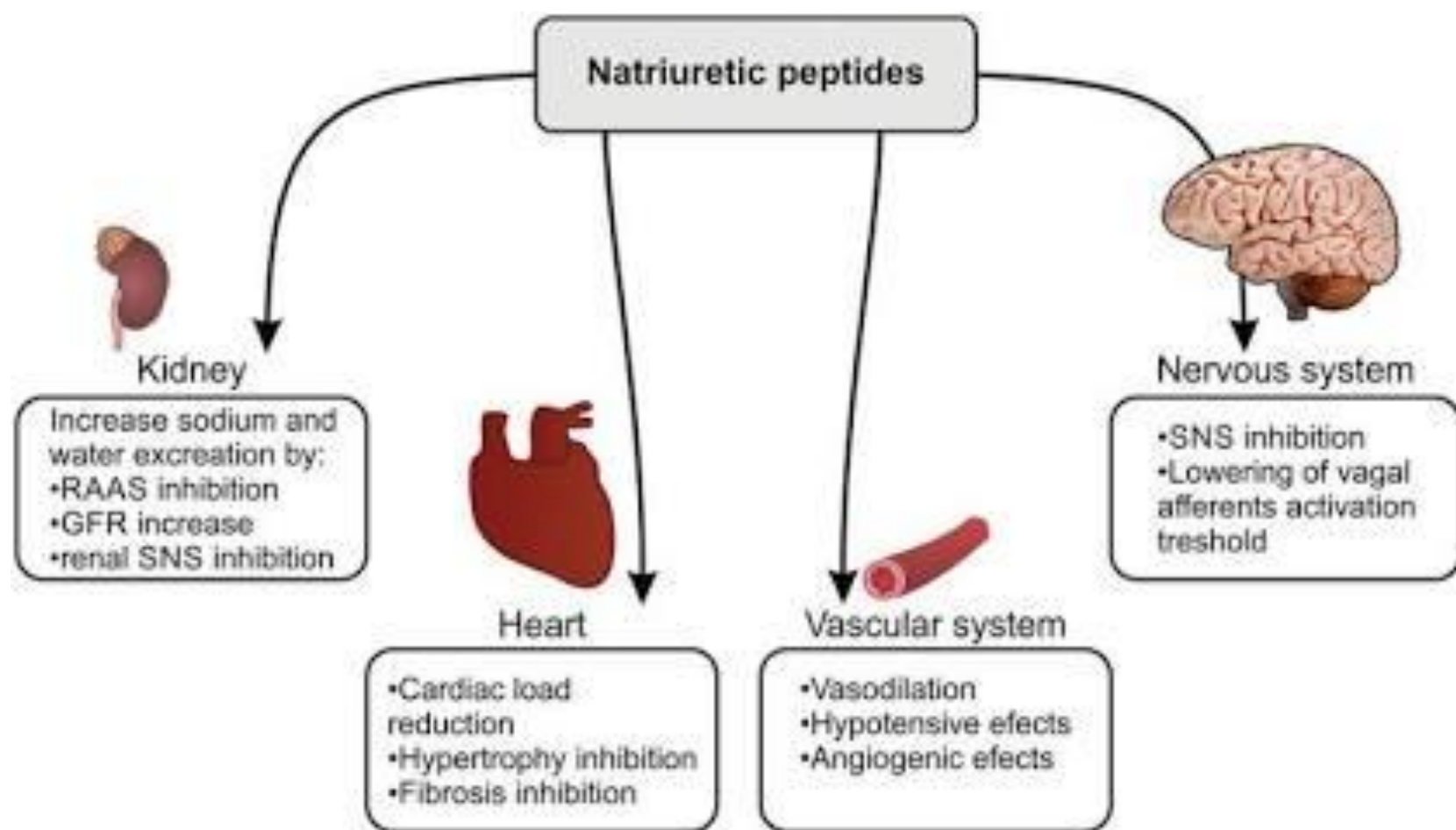
- HTN, atherosclerotic disease, or a narrow aortic valve opening increases arterial impedance (afterload), thereby increasing the workload of the heart
- HTN is a major factor in the development of both HFrEF and HFpEF

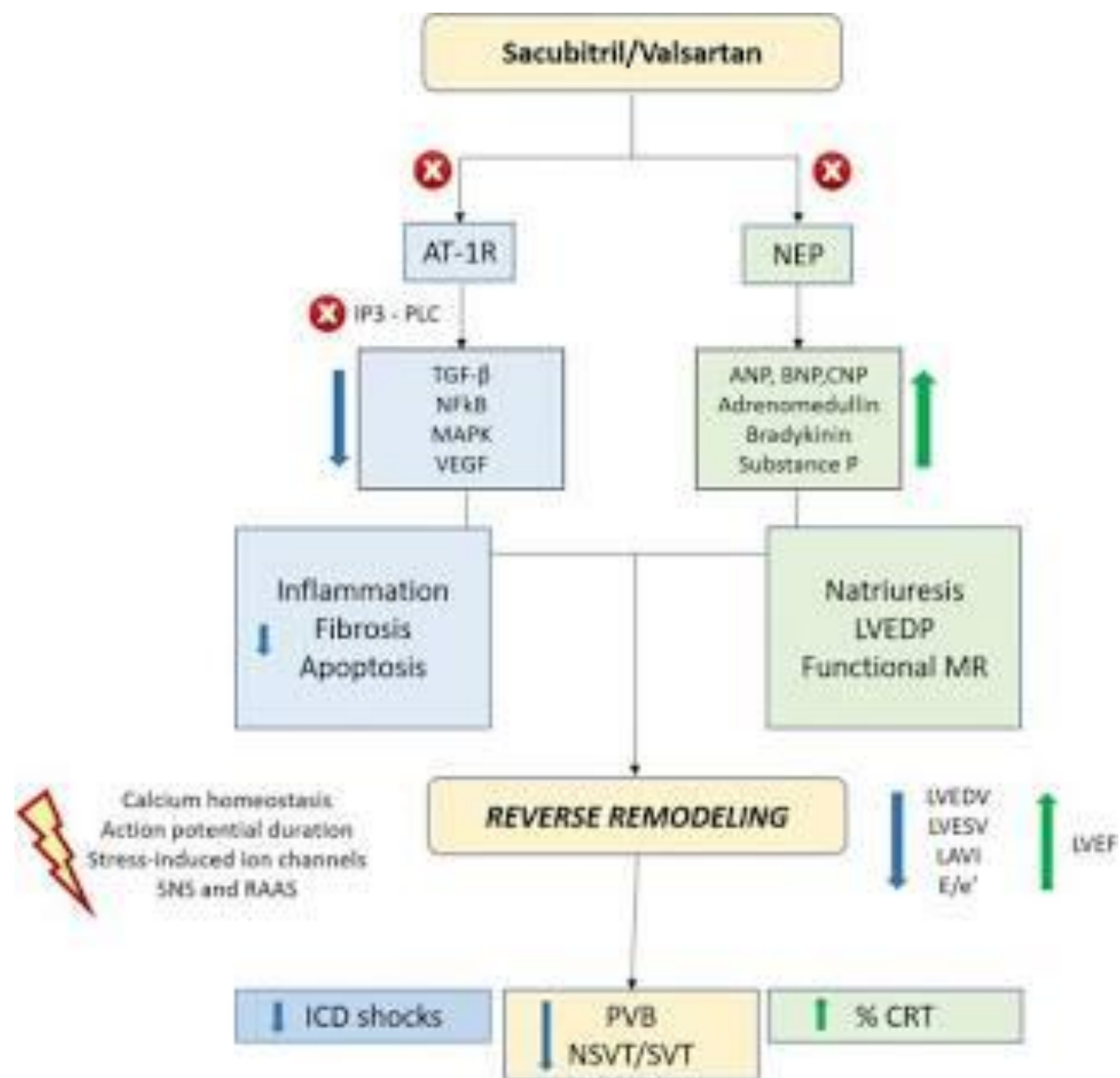
- **Cardiac contractility**

- ✓ myocardium's ability to develop force and shorten its fibers independent of preload or afterload
- ✓ decreased when myocardial fibers are diminished or poorly functioning (**cardiomyopathy, valvular HD, CAD, or after an MI**)
- ✓ A major factor for systolic HF
- ✓ BBs or anthracyclines precipitate HF by decreasing myocardial contractility

- **Heart rate**

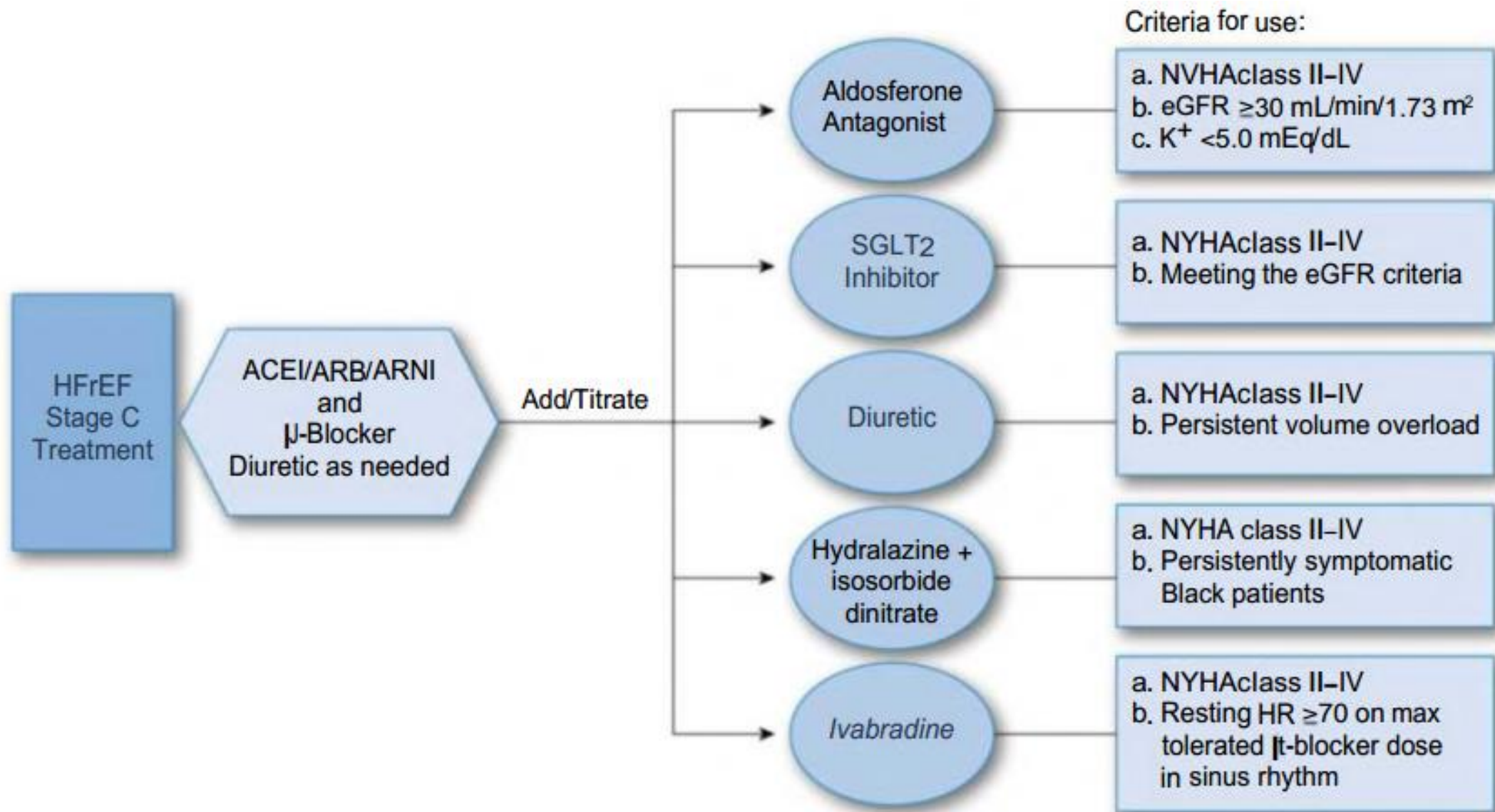
- ✓ ↑ed HR reflex mechanism to improve CO as EF declines
- ✓ The SNS is the major mediator of this response
- ✓ The workload and energy demands of a rapid HR ultimately place strain on the heart and can eventually worsen HF

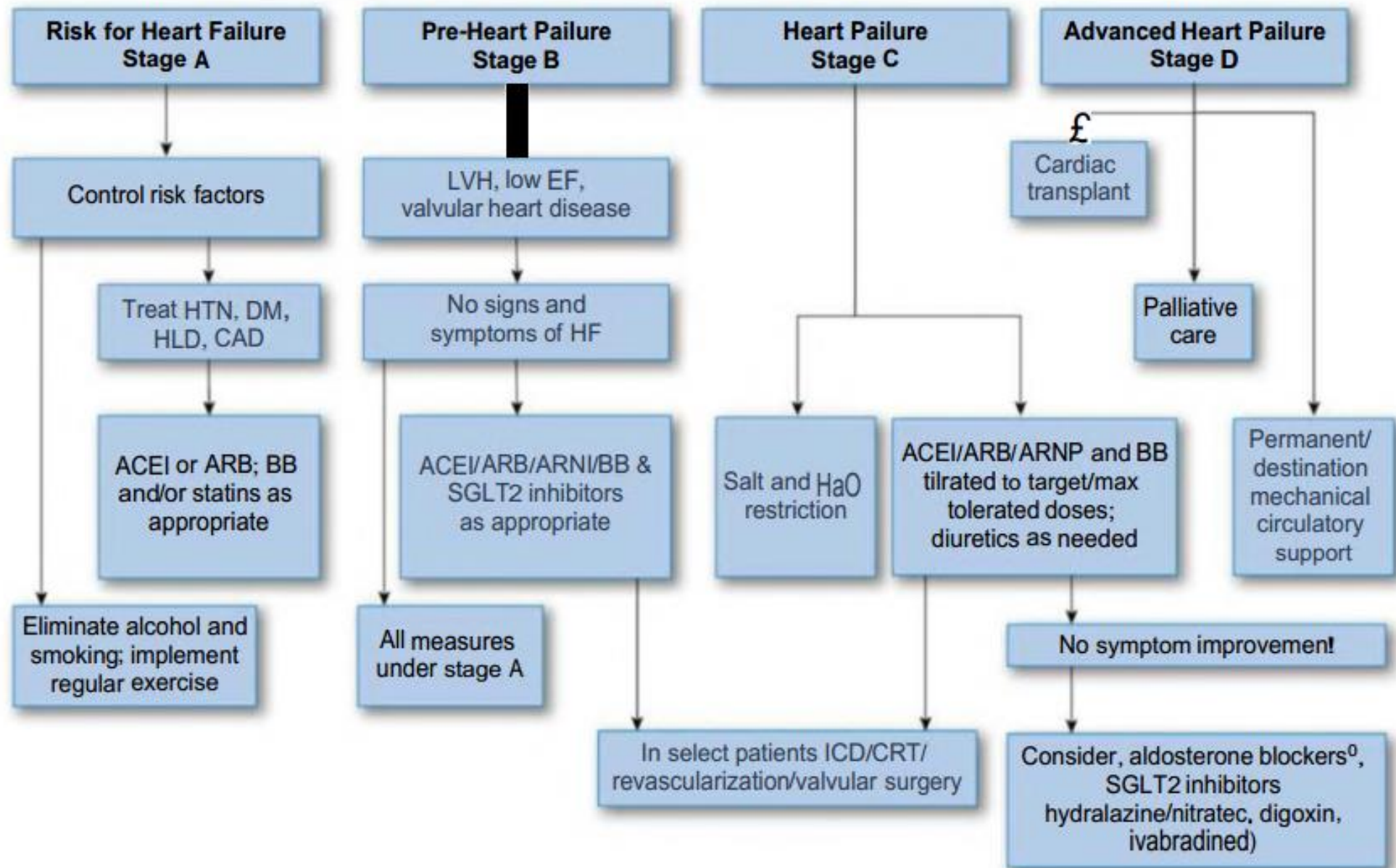




DRUG OR DRUG CLASS	MECHANISM OF ACTION	HEMODYNAMIC EFFECT	CLINICAL NOTES
Drugs with Proven Mortality Reduction			
ACE inhibitors	Inhibit AT II generation → ↓ AT _I receptor activation	Decreased afterload Decreased preload	May cause hyperkalemia
β-Antagonists	Competitive antagonists at β-adrenergic receptor → ↓ renin release	Decreased afterload Decreased preload	May be relatively contraindicated in severely decompensated heart failure
Spironolactone	Competitive antagonist at aldosterone receptor	Decreased preload	Mortality benefit may be independent of hemodynamic effects; may cause hyperkalemia

Drugs or Treatments Used to Improve Symptoms			
Na ⁺ /H ₂ O restriction	Decreases intravascular volume	Decreased preload	May help limit edema formation
Diuretics	Inhibit renal Na ⁺ reabsorption	Decreased preload	Furosemide most effective for treating congestive symptoms
Aquaretics	Competitive antagonists at vasopressin V ₂ receptor → ↓ renal aquaporin expression and membrane trafficking → ↓ free water reabsorption	Decreased preload	Increased output of solute-free urine; increased serum sodium
Digoxin	Inhibits Na ⁺ /K ⁺ ATPase → ↑ intracellular Ca ²⁺ → ↑ contractility	Increased contractility	Delays atrioventricular nodal conduction
Organic nitrates	Increase NO → venous smooth muscle relaxation → ↑ venous capacitance	Decreased preload	Reduces myocardial O ₂ demand
Dobutamine	Stimulates β-adrenergic receptors	Increased contractility (β ₁ effect) Decreased afterload (β ₂ effect)	Used in the acute setting only
Inamrinone, milrinone	Inhibit phosphodiesterase → ↑ β-adrenergic effect	Increased contractility Decreased afterload Decreased preload	Used in the acute setting only





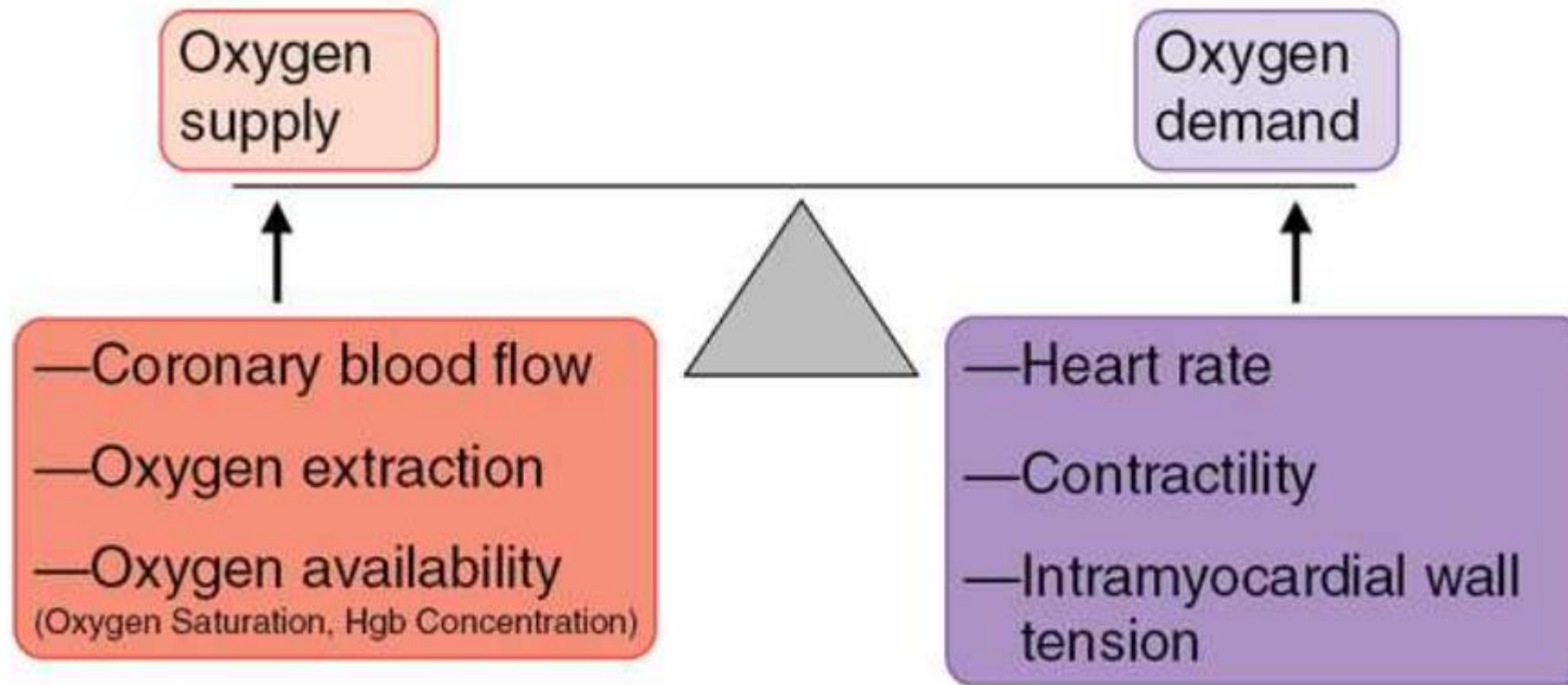
Ischaemic Heart Disease

❖ consequences of coronary atherosclerosis

- ✓ angina (chest pain caused by cardiac ischaemia)
- ✓ myocardial infarction

Angina

- occurs when the oxygen supply to the myocardium is insufficient for its needs.
- The pain in the **chest, arm and neck**
- brought on **by exertion, cold or excitement.**
- chemical factors released by ischaemic muscle are responsible



Determinants of myocardial oxygen supply and demand

- Oxygen extraction by heart cells is high (~70%–75%) even at rest
- Reaches plateaus at ~80%
- increased only modestly when the heart is stressed
- high oxygen demands must be met by increases in coronary blood flow

Atherosclerosis
A consequence of
high cholesterol

Normal artery

Outer layer —
Muscle layer —
Lining —

When the level of cholesterol in the bloodstream is normal, arterial walls remain smooth and slippery.

Clogged artery

LDL cholesterol —
Foam cell —
Plaque —

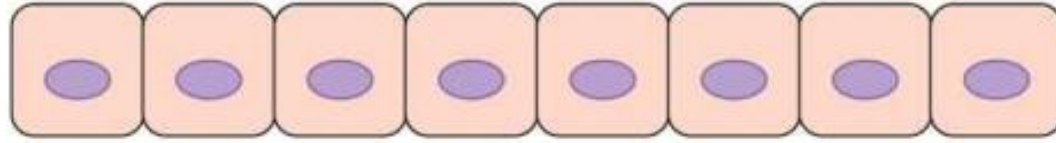
When cholesterol levels are high, excess cholesterol accumulates in the walls of arteries, reducing blood flow.

Traditional risk factors

- smoking
- Hypertension
- Hyperlipidemia
- Diabetes
- obesity
- producing oxidative stress within the vasculature
- progressive decreases in nitric oxide (NO) levels and endothelial dysfunction

Process of atherosclerosis

Endothelial Layer



- **Normal function**

- Substances produced
 - Nitric oxide (NO)
 - Prostaglandin I₂ (PGI₂)
- Main actions
 - Antithrombotic
 - Anti-inflammatory
 - Extracellular growth inhibition

- **Endothelial dysfunction**

- Substances produced
 - Angiotensin II
 - Endothelin
- Main actions
 - Prothrombotic
 - Proinflammatory
 - Extracellular growth facilitation

Stable angina

- Imbalance in myocardial oxygen supply and Demand
- produced by an increased demand on the heart
- usually caused by fixed narrowing(s) of the coronary vessels by atheroma
- narrowing of the aortic valve ('aortic stenosis')
 - reducing coronary blood flow even in the absence of coronary artery narrowing

Symptomatic therapy

- reducing cardiac work
- **organic nitrates** (sublingual nitroglycerin)-add on therapy on anti-ischemic drugs
- **β -adrenoceptor antagonists** and/or
- **calcium antagonists** (LA-monotherapy, with BBs)
- together with treatment of the underlying atheromatous disease (antiplatelet drug, aspirin, clopidogrel)

- restores the appropriate balance of myocardial oxygen supply and demand (**β-blockers, calcium channel blockers, and long-acting nitrates**)
- interrupts the negative consequences of myocardial ischemia on myocardial energy consumption (**ranolazine**)

Calcium Channel Blockers in Anginal Syndromes

Drug Name	FDA Approved*	Usual Dose for Chronic Stable Angina ⁶	Product Availability
Dihydropyridines			
Amlodipine	Angina, hypertension	2.5–10 mg every day	2.5, 5, 10 mg tab
Felodipine	Hypertension	5–20 mg every day	5, 10 mg ER tab
Isradipine	Hypertension	2.5–10 mg bid	2.5, 5 mg IR cap
Nifedipine	Angina (IR only), hypertension	5–10 mg every day	5, 10 mg CR tab
		20–40 mg tid	20, 30 mg IR cap
Nifedipine	Angina, hypertension	30–60 mg bid	30, 45, 60 mg SR cap
		10–30 mg tid	10, 20 mg IR cap
Nifedipine	Hypertension	30–180 mg every day	30, 60, 90 mg ER tab
		20–60 mg every day	10, 20, 30, 40 mg ER tab
Diphenylalkylamines			
Verapamil	Angina, hypertension, SVT	30–120 mg tid/qid	40, 80, 120 mg IR tab
		120–240 mg bid	120, 180, 240 mg SR tab
		120–480 mg every HS	180, 240 mg DR, ER tab
			120, 180, 240, 360 mg ER cap
			100, 200, 300 mg DR, ER tab
Benzothiazepines			
Diltiazem	Angina, hypertension, SVT	30–120 mg tid/qid	30, 60, 90, 120 mg IR tab
		60–180 mg bid	60, 90, 120, 180 mg SR cap
		120–480 mg every day	120, 180, 240, 300, 360 mg cap
			120, 180, 240 mg ER cap
			120, 180, 240, 300, 360, 420 mg ER cap

- more potent peripheral vasodilators
- reflex tachycardia

- moderate peripheral vasodilators
- potent coronary artery vasodilators

- increase in myocardial oxygen supply (VD), reduce myocardial oxygen demand (lowers BP)-both

- compromised left ventricular (LV) function or bradycardia at baseline.
 - amlodipine and felodipine safe to use
- **used cautiously:** with cyclosporine, carbamazepine, lithium, amiodarone, and digoxin
 - Avoid Non-DHP CCB
 - Negative Chronotropy
 - Negative Inotropy
 - Reduced Ejection Fraction (HFrEF)
 - bradycardia, hypotension, and AV block

Calcium Channel Blocker Hemodynamic and Electrophysiologic Profile

Effect	Dihydropyridine Derivatives ^a	Diltiazem	Verapamil
Peripheral vasodilation ^b	+++	++	++
Coronary vasodilation ^b	+++	+++	++
Negative Inotropes ^c	±	++	+++
AV node suppression ^a	±	+	++
Heart rate	Increase (reflex)	Decrease or unchanged	Decrease or unchanged
Pharmacokinetics ^d			
Dose/mg ^e			
<i>Side Effects</i>			
Nausea, vomiting	+ (most)	±/1	±
Constipation	Not observed	±	±
Hypotension, dizziness ^f	++	+	+
Flushing, headache	++	+	+
Bradycardia, HF symptoms	±	+	++
ReFlex tachycardia, angina	V	Not observed	Not observed
Peripheral edema	+	±	±

- ***RANOLAZINE***

- inhibits the **late sodium current**, thereby reducing intracellular sodium
- prevents ischemia-induced contractile dysfunction and delays the onset of angina.
- Extensively metabolized in the liver through both cytochrome P-450 enzymes CYP3A4 and CYP2D6 (ketoconazole or rifampin)
- C/I: significant hepatic dysfunction
- A/E: headache, constipation, dizziness, and nausea

Considerations for the Use of Ranolazine in Patients With Chronic Stable Angina

Clinical Issue	Recommended Management Strategy
Renal insufficiency	Ranolazine plasma levels may increase up to 50%. Caution with dose titration to maximal recommended dose.
Hepatic insufficiency	Ranolazine is contraindicated in patients with clinically significant hepatic impairment.
<i>Drug Interactions: Effects on Ranolazine</i>	
Strong CYP3A4 inhibitors	Plasma concentrations of ranolazine are significantly elevated when combined with potent inhibitors of CYP3A4. Ranolazine is contraindicated in patients receiving strong CYP3A4 inhibitors (ketoconazole, clarithromycin, nelfinavir, etc).
Moderate CYP3A4 inhibitors	Limit the dose of ranolazine to 500 mg twice daily in patients receiving moderate inhibitors of CYP3A4 (diltiazem, verapamil, erythromycin, fluconazole, etc).
CYP3A4 inducers	Coadministration of ranolazine with CYP3A4 inducers is contraindicated and should be avoided.
P-glycoprotein inhibitors	Caution should be exercised when coadministering ranolazine with P-glycoprotein inhibitors, and the dose of ranolazine may need to be lowered based on clinical response.

Drug Interactions: Effects on Other Medications

Simvastatin	Plasma levels of simvastatin are increased twofold with coadministration with ranolazine through CYP3A4 inhibition by ranolazine; limit simvastatin dose of 20 mg daily when used with ranolazine; closely monitor for adverse effects (eg, myositis) from simvastatin.
Digoxin	Ranolazine coadministration increases plasma concentrations of digoxin by 1.5 times. Adjust dose of digoxin accordingly to maintain desired therapeutic level and response.
Metformin	Limit the dose of metformin to 1700 mg daily when coadministered with ranolazine 1000 mg twice daily.
CYP2D6 substrates	Ranolazine can inhibit the activity of CYP2D6, and plasma concentrations of 2D6 substrates (β -blockers, tricyclic antidepressants, antipsychotics) may be increased and lower doses of these agents may be required.
QT prolongation	Caution is recommended if the patient is on other QT-prolonging drugs, or has QT prolongation as baseline.

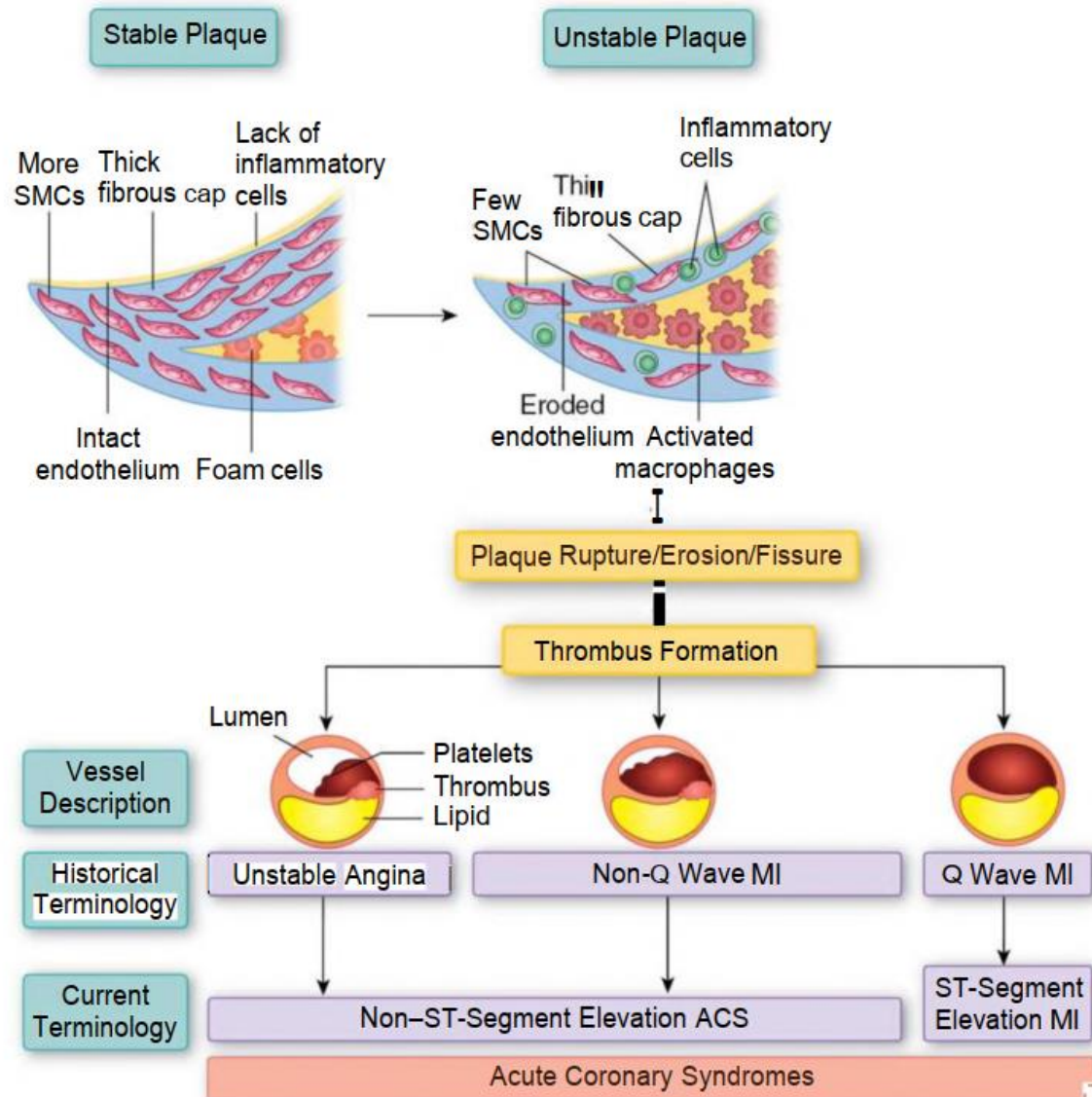
Vasospastic angina

- hallmark of variant angina is ST-segment elevation
 - denotes rapid and complete occlusion of the coronary artery
- reversible narrowing
- caused by increased coronary vascular resistance
- coronary artery vasodilators (e.g. organic nitrates, calcium antagonists)
- CCBs:
 - alone or in combination with long-acting nitrates for nocturnal vasospastic angina
 - equally effective in preventing Prinzmetal variant angina
 - Intrinsically long-acting or sustained-release forms are preferred

- organic nitrates
- Avoid BBs

Acute Coronary Syndrome

- includes
 - unstable angina (UA) or
 - acute myocardial infarction (MI)
 - ST-segment elevation myocardial infarction (STEMI) and non–ST-segment elevation myocardial infarction (NSTEMI)
- **Treatment goals:** to alleviate ischemic symptoms, restore blood flow to the infarct-related artery, arrest infarct expansion, minimize adverse outcomes, and prevent mortality



Thrombus formation and acute coronary syndrome definitions

- initial therapies may include (for both)
 - ✓ Oxygen
 - ✓ nitroglycerin (NTG)
 - ✓ antiplatelet agents
 - ✓ Anticoagulants
 - ✓ β -blocker
 - ✓ angiotensin-converting enzyme (ACE) inhibitor or
 - ✓ angiotensin receptor blocker (ARB)
 - ✓ morphine sulfate

STEMI

- to restore coronary blood flow by administering a fibrinolytic or performing percutaneous coronary intervention (PCI)
- patients should receive aspirin in addition to a P2Y₁₂ inhibitor

Acute Supportive Care Medications Initiated During the Initial 24 Hours of ACS Treatment

Drug	Indication (Class of Recommendation) ^a	Contraindication/Caution	Dose/Administration	Adverse Effects
Morphine	Refractory pain (IIb)	Known hypersensitivity Hypotension Bradycardia Lethargic or moribund patient	STEMI: 4-8 mg IV × 1 (lower dose in older patients), then 2-8 mg IV every 5-15 minutes PRN NSTEMI-ACS: 1-5 mg IV every 5-30 minutes PRN	Constipation, nausea, vomiting, hypotension, respiratory depression
Oxygen	Oxygen saturation <90% [0.90] (I)	Chronic obstructive pulmonary disease Carbon dioxide retention	2-4 L/min, increasing rate and/or changing to face mask PRN	Increased coronary vascular resistance, decreased coronary blood flow, increased mortality
Nitroglycerin	Angina (I) Uncontrolled hypertension (I) Acute heart failure (I)	SBP less than 90 mm Hg or greater than 30 mm Hg below baseline Avoid if recent PDE ₅ inhibitor use: • Avanafil: within 12 hours • Sildenafil: within 24 hours • Vardenafil: within 24 hours • Tadalafil: within 48 hours Use with caution if RV infarct suspected Avoid abrupt cessation of IV nitroglycerin; wean gradually	SL: 0.3-0.4 mg every 5 minutes, up to 3 doses PRN IV: 10 mcg/min titrated to symptom relief and desired blood pressure	Flushing, headache, hypotension, tachycardia

β-Blockers	All patients without contraindications (I) <i>Associated with mortality reduction, especially in patients with HFrEF</i>	Signs of heart failure Low cardiac output state Risk factors for cardiogenic shock: • Age 70 years or greater • SBP less than 120 mm Hg • Sinus tachycardia (HR greater than 110 bpm) • Sinus bradycardia (HR less than 60 bpm) • Killip class III • Prolonged time from symptom onset High-grade AV block Active asthma or reactive airway disease	Carvedilol 6.25 mg twice daily; target dose is 25 mg twice daily as tolerated Metoprolol • Oral: 25-50 mg every 6-12 hours for 2-3 days, then once (metoprolol succinate) or twice daily (metoprolol tartrate); target dose is 200 mg daily • IV: 5 mg every 5 min as tolerated up to 3 doses, titrated to BP and HR; should only be considered if BP is uncontrolled or refractory symptoms Continue indefinitely in patients with concomitant HFrEF Other β-blockers may be considered; in patients with HFrEF, use either metoprolol succinate, carvedilol, or bisoprolol	Hypotension, heart failure, bradycardia, cardiogenic shock, AV block, exacerbation of asthma or reactive airway disease
Calcium channel blockers	Angina, normal LVEF, and contraindication or intolerance to β-blocker (I) Angina refractory to β-blocker and normal LVEF (I) Coronary vasospasm (I)	Signs of heart failure Low cardiac output state Risk factors for cardiogenic shock: • Age 70 years or greater • SBP less than 120 mm Hg, sinus tachycardia (HR greater than 110 bpm) • Sinus bradycardia (HR less than 60 bpm) • Killip class III • Prolonged time from symptom onset High-grade AV block	Diltiazem 120-360 mg/day orally ^b Verapamil 240-480 mg/day orally ^b Amlodipine 5-10 mg orally once daily Nicardipine 60-120 mg/day orally ^b Nifedipine ER 30-120 mg orally once daily	Hypotension, diltiazem and verapamil: heart failure, cardiogenic shock, bradycardia, AV block

Treatment Strategies in STEMI